

Note:
- Instead of using array maybe use classes and attributes
--> should be more performance optimized

--> class: dataset
--> attributes: e.g. dataset.accelerationX1

- Logging optional depending on uC performance

- All Status changes should be communicated with telemetry

- Instead of Booleans use Status byte and check for status byte
Statusbyte structure (should be send to telemetry at the beginning of each loop):

- 1 byte (char)
- States:
 - 0000 0000 (0): Pad Idle
 - 0000 0001 (1): Armed
 - 0000 0010 (2): Liftoff detected
 - 0000 0011 (3): Booster Burnout/Coasting
 - 0000 0100 (4): Apogee detected
 - 0000 0101 (5): Pyro1 signal sent (drogue)
 - 0000 0110 (6): Pyro2 signal sent (drogue)
 - 0000 0111 (7): Pyro3 signal sent (drogue)
 - 0000 1000 (8): Pyro4 signal sent (drogue)
 - 0000 1001 (9): Drogue deployment detected
 - 0000 1010 (10): Pyro1 signal sent (main)
 - 0000 1011 (11): Pyro2 signal sent (main)
 - 0000 1100 (12): Pyro3 signal sent (main)
 - 0000 1101 (13): Pyro4 signal sent (main)
 - 0000 1110 (14): Main deployment detected
 - 0000 1111 (15): Landed
- maybe safe two pyro statuses per parachute (at most two channels per parachute)
 - able to add continuity checks

